Upper-body evaluation — loading-gate tests

Shoulder · elbow · wrist · low back — ranges, control and symmetry · 2 sessions (~75 + ~25 min)

Muscle group

■ Shoulder / rotator cuff ■ Elbow (biceps, triceps) ■ Wrist / forearm ■ Low back / trunk

Before you start — what this is, and how the phone is used

What: tests that decide when each upper-body joint may move up one loading stage. Order: A, B, C, then D.
When: a rested day, no upper-body training or badminton the day before. Warm up 5 min (walk or bike).
Phone = camera only: on the tripod ~2 m away, level, lens at the height the card names; never in your hand.
Beat: a metronome on a second device (PC, tablet or speaker), loud enough for the video to record it.
iPhone (iOS 18 or later): turn on Settings → Camera → Record Sound → Allow Audio Playback.
Angles: later, on the PC, in Kinovea (free): open the video, pause on the frame, pick the card's tool, drag its centre onto the joint, note the number (over 180°: 360 minus it), snip the frame (Win+Shift+S).
Pictures move when this file opens in Edge or Chrome; elsewhere they show the start (dark) and end (faint).
Kit: dumbbells 2 × 1.5 kg, 2.5 kg, 1 kg, a curl weight · hex dumbbell handles · a band · bench · wall · mat
2 pens · tape · tape measure · tripod and phone · a second device for the beat · Kinovea on the PC

A — Session 1: Ranges (~20 min · no weights · your own effort only, nothing pushes further)

1 Shoulder raise range — each arm

3 tries per arm
Tripod · phone

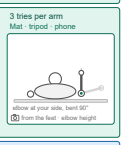
Set-up: Stand side-on to the camera, 2 m away, lens at shoulder height, the tested arm nearest the camera, thumb up.
Do: Raise the straight arm forward and up as far as it goes by itself; hold 2 s; lower. 3 tries, then turn round for the other arm.
Read: Kinovea, top frame: Angle tool, centre on the shoulder, one leg to the hip, the other to the elbow.
Write down: The best of 3, per arm. Your training limit is ~75% of it.
How to: Kinovea angle tools ↗



2 Shoulder turn-out range, lying — each arm

3 tries per arm
Mat · tripod · phone

Set-up: Lie on your back, upper arm against your side, elbow bent 90°, forearm pointing at the ceiling (0°). Camera at your feet, lens at elbow height, looking along the arm.
Do: Turn the forearm out toward the floor as far as it goes by itself, the upper arm still against your side; hold 2 s; back up; 3 tries.
Read: Kinovea, end frame: Angle-to-vertical tool along the forearm, elbow to wrist.
Write down: The best of 3, per arm. Training limit: 45° or ~75% of it, whichever is smaller.
How to: Kinovea angle tools ↗



3 Elbow range — each arm

3 tries per arm
Tripod · phone

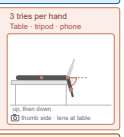
Set-up: Stand side-on, the tested arm nearest the camera, hanging, palm forward. Lens at elbow height, 2 m away.
Do: Straighten the elbow fully, hold 2 s; then bend it fully, hold 2 s. 3 tries.
Read: Kinovea, both end frames: Angle tool, centre on the elbow, legs to the shoulder and the wrist.
Write down: Both numbers per arm, and whether the elbow bows past straight (yes/no). Training stops 20–30% of the arc short of straight.
How to: Kinovea angle tools ↗



4 Wrist range — each hand

3 tries per hand
Table · tripod · phone

Set-up: Forearm flat on a table, palm down, hand past the edge. Camera at the thumb side, lens at table height.
Do: Bend the hand up as far as it goes by itself, hold 2 s; then down, hold 2 s. 3 tries.
Read: Kinovea: Angle tool, centre on the wrist, one leg along the forearm, the other to the knuckles.
Write down: The up and the down number, per hand. Training limit: 70–80% of each.
How to: Kinovea angle tools ↗



5 Low-back bend range — forward and back

3 tries each way
2 pens · tape · tripod

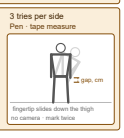
Set-up: Tape one pen flat on your spine at the two low-back dimples, a second pen 15 cm higher. Stand side-on, knees straight, lens at waist height.
Do: Stand tall 2 s; bend forward as far as is comfortable, hold 2 s; stand up; lean back, hold 2 s. 3 tries each way.
Read: Kinovea, on the standing, forward and back frames: Angle-to-horizontal on each pen.
Write down: Both pen numbers on each frame; Claude works out the range. Limit: 70–80% of it, never past ~36–42° forward or ~12–15° back.
How to: Kinovea angle tools ↗



6 Low-back side-bend — each side

3 tries per side
Pen · tape measure

Set-up: Stand tall, arms by your sides, feet hip-width apart. Mark where your middle fingertip touches the thigh.
Do: Slide the hand straight down the side of the thigh as far as is comfortable, without leaning forward or back; mark again. 3 tries per side.
Write down: The gap between the marks (cm), each side. Training limit: 70–80% of it. Twisting has no number: it is judged on the control-test videos.



B — Session 1: Control tests (~35 min · each clean or not · film every one)

7 Hip hinge (waiter's bow)

3 slow reps
Tripod · phone

Set-up: Stand side-on, hands on hips, knees soft. Lens at hip height.
Do: Bow forward from the hips to about 60°, back flat, then stand up. 3 slow reps.
Fail if: The low back rounds, or the hips stop bending before 50° without the back moving.
Write down: Failed reps (a test fails at 2 of 3).
Photo: Luomajoki 2007, Fig 1 ↗ Video: IFMK Brest, all six tests (French) ↗



8 Pelvic tuck (pelvic tilt)

3 slow reps
Tripod · phone

Set-up: Stand side-on, relaxed. Lens at hip height.
Do: Squeeze the glutes and tuck the pelvis under, so the low back flattens, while the upper back stays still; relax. 3 slow reps.
Fail if: The pelvis does not tilt, the low back arches instead, the glutes stay off, or the upper back rounds.
Write down: Failed reps.
Photo: Luomajoki 2007, Fig 4 ↗ Video: IFMK Brest, all six tests (French) ↗



9 Seated knee straighten — each leg

3 reps per leg
Bench · tripod

Set-up: Sit tall on a high bench, feet off the floor, low back in its normal small arch. Lens at seat height, side-on.
Do: Straighten one knee slowly until it is 30–50° short of straight, then lower. 3 reps per leg.
Fail if: The low back rounds (slumps) as the knee straightens.
Write down: Failed reps, per leg.
Photo: Luomajoki 2007, Fig 2 ↗ Video: IFMK Brest, all six tests (French) ↗



10 Rock forward on all fours

3 slow reps
Mat · handles · tripod

Set-up: On hands and knees, fists or handles under the shoulders, knees under the hips, back flat. Lens ~30 cm up, side-on.
Do: Rock the body forward, shoulders past the hands, and back. 3 slow reps.
Fail if: The low back sags (arches) as you rock forward.
Write down: Failed reps.
Photo: Luomajoki 2007, Fig 6 ↗ Video: IFMK Brest, all six tests (French) ↗



11 Rock back on all fours

3 slow reps
Mat · handles · tripod

Set-up: As card 10.
Do: Rock the hips back toward the heels until the hips are bent about 120°, back flat; return. 3 slow reps.
Fail if: The low back rounds before the hips reach about 120°.
Write down: Failed reps.
Photo: Luomajoki 2007, Fig 3 ↗ Video: IFMK Brest, all six tests (French) ↗



12 Face-down knee bend — each leg

3 reps per leg
Mat · tripod

Set-up: Lie face down, forehead on your hands, legs straight. Lens at floor level, side-on.
Do: Bend one knee until the shin is upright (90°) or more, then lower. 3 reps per leg.
Fail if: The pelvis lifts or tips, or the low back arches, as the knee bends.
Write down: Failed reps, per leg. Tests 7–12 are clean with at most 1 failed test and no pain.
Photo: Luomajoki 2007, Fig 5 ↗ Video: IFMK Brest, all six tests (French) ↗



13 Shoulder-blade check

8–10 + 8–10, twice
2 × 1.5 kg · tripod

Set-up: Shirt off, back to the camera 2 m away, lens at shoulder height. Metronome 60: 2 beats up, 2 down.
Do: First without weights, then with 1.5 kg in each hand: raise both straight arms forward, thumbs up, as high as is comfortable, 8–10 times; then out to the sides, 8–10 times.
Fail if: On the video, a blade's inner edge lifts off the ribs (2.5 cm or more), or the blade jerks, hitches up early or drops fast, on 3 of 5 reps or more; worse with weights than without; any slipping or pain.
Write down: Per side and per raise: normal, subtle (mild, not every rep) or obvious.
Photo: McClure 2009, Fig 2 (normal vs obvious) ↗



14 Kneeling push-ups on handles

10 reps × 2 sets
Hex DB handles · tripod

Set-up: Kneel, hands on hex-dumbbell handles under the shoulders, body straight from knees to head. Lens at floor level, side-on; film one set from each side. Metronome 60.
Do: Down for 3 beats until the upper arms are level with the body, up for 2, hold 2 at the top with soft elbows. 10 reps.
Fail if: An elbow goes straight (less than ~15° bent on the top frame), snaps, clunks or drifts into lockout; any pain.
Write down: Clean or not, per elbow.
Video: Get Exercise Confident, kneeling push-up ↗



15 One-arm hold on all fours — each arm

30 s per arm
Mat · handles · tripod

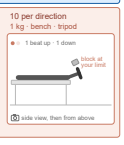
Set-up: On hands and knees as card 10, lens side-on, the supporting arm nearest the camera.
Do: Lift the other hand off the floor and hold 30 s, the supporting elbow slightly bent. Then switch arms.
Fail if: As card 14: the elbow goes straight, snaps, clunks or drifts into lockout; any pain.
Write down: Clean or not, per elbow.
Video: NASM, quadruped arm raise ↗



16 Wrist control set, 1 kg — each hand

10 per direction
1 kg · bench · tripod

Set-up: Forearm supported on a bench, hand past the edge; a book or block at your card-4 limit stops the hand. Film from the side, then from above. Metronome 60: up on one beat, down on the next.
Do: 10 up · 10 down · 10 side to side (palm down, thumb side to little-finger side) · 10 palm-down to palm-up turns. Side to side and turns stay small.
Fail if: A clunk or catch, the wrist gives way, the hand passes the stop, a beat is missed, or any pain.
Write down: Clean or not, per hand.
Video: AskDoctorJo, wrist extension ↗ turns ↗

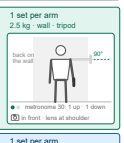


C — Session 1: Symmetry tests (~20 min · each arm to technical failure · ≥3 min between sides)

17 Side-raise endurance, 2.5 kg — each arm

1 set per arm
2.5 kg · wall · tripod

Set-up: Back and head against a wall, feet shoulder-width. Camera in front, 2 m, lens at shoulder height. Metronome 30: up on one beat, down on the next (2 s each).
Do: Raise the arm to shoulder height, about 30° in front of straight out to the side, elbow slightly bent, thumb pointing forward and level; lower. Keep going.
Stop when: A rep misses shoulder height, misses the beat or drifts out of line.
Write down: Reps before the first miss, per arm.
Photo: Ford 2026, Fig 1 (step by step) ↗



18 Curl endurance, one arm — each arm

1 set per arm
Curl weight · bench

Set-up: Sit tall, back supported, one dumbbell that gives your stronger arm 12–20 reps; the same weight every time. Camera side-on. Metronome 60: 2 beats up, 2 down.
Do: Curl up and lower; the bottom stops just short of straight; no swinging.
Stop when: A rep misses the beat or the range.
Write down: Reps per arm, and the weight (kg).
Video: OPEX, seated one-arm curl ↗



19 Band wrist-lift endurance — each hand

1 set per hand
Band · bench · tripod

Set-up: Forearm supported, palm down, hand past the edge, the band from under your foot to the hand; the same band and spot every time. Camera side-on. Metronome 60: 2 beats up, 2 down.
Do: Lift the back of the hand and lower it, inside your card-4 limits.
Stop when: A rep misses the beat or the range.
Write down: Reps per hand, and which band.
Video: Baptist Health, band wrist extension ↗



D — Session 2: 2–7 days later (~25 min · symmetry tests only)

20 Repeat 17–19 — other side first

1 set per arm
As session 1

Set-up: 2–7 days after session 1: the same warm-up, weights, band, beat and set-up. Start with the other side.
Do: Tests 17, 18 and 19 once more.
Score: Mean of both sessions, weaker ÷ stronger: shoulder ≥80%, elbow and wrist ≥85% passes.
Example: Side raise left 18 and 17, right 21 and 22: 17.5 ÷ 21.5 = 81%, a pass.



Step-gate log — after every training session (exit criterion 1)

Per joint: pain ≤5/10 during and after · back to normal the next morning · no swelling or warmth within 24 h · no "loose joint" feeling that day.
Two clean weeks in a row are needed; a fail is a flare and restarts the count.

Every test — rules

- These are tests, not training sets: follow each card's set-up exactly
- Stop at pain, a clunk or a slipping feeling — that test is not clean this round
- Same weight, band, set-up and beat every time; write down the curl weight and the band
- Film every test and judge it on the paused video, not by feel
- Ranges: your own effort only — no hand, strap or body weight pushes the joint further
- No stretching before the range tests

What the results unlock — the same tests at every stage step

A joint moves up one loading stage when all three hold: 2 clean step-gate weeks, a clean control test and a passed symmetry test. Passing now opens stage 2 only.
For 2→3: train ≥2 clean weeks at stage 2, then run this file again; the same for 3→4.
The ranges (A) set your training limits at every stage; redo them after a flare.
The low back has no symmetry test (not a paired joint) · a flare drops a joint back one stage.
Rules behind this file: Stage 0 §2.5–2.6 · R7 · R14 · decisions D-63…D-79 (gymspec folder).

Send back to Claude — results per joint

Shoulder: raise ° L/R · turn-out ° L/R · blade rating L/R (forward, side) · side-raise reps L/R ×2
Elbow: straight and bent ° L/R · past straight? · tests 14–15 clean? · curl reps L/R ×2 · curl kg
Wrist: up and down ° L/R · test 16 clean? · band reps L/R ×2 · which band
Low back: pen numbers on the 3 frames · side-bend cm L/R · failed tests __ of 6
All: the Kinovea snips · step-gate clean weeks so far, per joint